

Fourier Analysis on Schwartz Space

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This note mainly covers Chapter 3.3 of Taylor. For a rigorous treatment, one may also read Chapter 8.3 of Folland.

1 L^1 theory

Definition 1.1 (Fourier transform). *Let $f \in L^1(\mathbb{R}^n)$, the Fourier transform of f is defined by:*

$$\mathcal{F}f(\xi) := \hat{f}(\xi) := \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx$$

The integral is well-defined since

$$|\mathcal{F}f(\xi)| \leq \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} |f(x)| |e^{-ix \cdot \xi}| dx = \frac{1}{(2\pi)^{n/2}} \|f\|_{L^1} < \infty.$$

Remark 1.2. *The definition of Fourier transform may vary. Fix one textbook and you'll be fine.*

Thus $\mathcal{F} : L^1(\mathbb{R}^n) \rightarrow L^\infty(\mathbb{R}^n)$.

In fact, we have a much stronger result.

Theorem 1.3 (Riemann-Lebesgue lemma). $\mathcal{F} : L^1(\mathbb{R}^n) \rightarrow C_0(\mathbb{R}^n) := \{f \in C(\mathbb{R}^n) \mid \lim_{|x| \rightarrow \infty} f(x) = 0\}$.

Proof. Folland, Theorem 8.22(f). □

We prepare a lemma for the main theorem.

Lemma 1.4 (Borrowing hat). *If $f, g \in L^1$, then*

$$\int_{\mathbb{R}^n} \hat{f} g = \int_{\mathbb{R}^n} f \hat{g}.$$

Proof. Fubini. □

2 Schwartz space

We will define a good space to do Fourier analysis.

Definition 2.1 (Schwartz space).

$$\mathcal{S}(\mathbb{R}^n) := \{f \in C^\infty(\mathbb{R}^n; \mathbb{C}) \mid \sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta f(x)| < \infty \forall \alpha, \beta\}$$

$\mathcal{S}$ has good algebraic properties. By triangle inequality, it's clear that $\mathcal{S}$ is a $\mathbb{C}$ -vector space. Moreover,

Proposition 2.2. *If $f, g \in \mathcal{S}$, then (exercise)*

(i) $f \cdot g \in \mathcal{S}$;

(ii) $x^\alpha f \in \mathcal{S}$ and $D^\beta f \in \mathcal{S}$ for all α, β ;

(iii) $f * g \in \mathcal{S}$.

Proposition 2.3 (Good integrability). *If $f \in \mathcal{S}$, then $D^\alpha f \in L^p$, $\forall 1 \leq p \leq \infty$.*

Proof. Divide the integral into two parts: $|x| \leq 1$ and $|x| > 1$. If $|x| \leq 1$, then $D^\alpha f$ is bounded as a continuous function. If $|x| > 1$, then $|D^\alpha f| = |x|^\beta |D^\alpha f| |x|^{-\beta} \leq C_{\beta, \alpha} |x|^{-\beta} \in L^p(x : |x| > 1)$ if $|\beta| \gg 0$. $\square$

Corollary 2.4. *$\mathcal{S}$ is dense in L^p , $1 \leq p < \infty$.*

Proof. $C_c^\infty \subset \mathcal{S} \subset L^p$, and C_c^∞ is dense in L^p , $1 \leq p < \infty$ (cf. Folland Proposition 8.17). $\square$

Example 2.5. Fix $a > 0$, then the function $f(x) = e^{-a|x|^2} \in \mathcal{S}$, satisfying

$$\int_{\mathbb{R}^n} e^{-a|x|^2} dx = \left(\frac{\pi}{a}\right)^{n/2}.$$

Theorem 2.6 (Commutator relation). *If $f \in \mathcal{S}$, then $\forall \alpha, \beta$,*

$$(i) \quad \mathcal{F}(D^\alpha f)(\xi) = \xi^\alpha \mathcal{F}(f)(\xi),$$

$$(ii) \quad D^\beta(\mathcal{F}(f))(\xi) = (-1)^{|\beta|} \mathcal{F}(x^\beta f)(\xi).$$

Equivalently,

$$(i) \quad \mathcal{F} \circ D^\alpha = M^\alpha \circ \mathcal{F},$$

$$(ii) \quad D^\beta \circ \mathcal{F} = (-1)^{|\beta|} \mathcal{F} \circ M^\beta,$$

where the first M^α is multiplication by ξ^α , and the second M^β is multiplication by x^β .

Proof. We compute from left to right. (i) is given by integration by parts and $f \in \mathcal{S} \subset C_0$, so that the boundary term vanishes. (ii) is given by differentiation under the integral sign. $\square$

Corollary 2.7.

If $f \in \mathcal{S}$, then $\mathcal{F}f \in \mathcal{S}$.

Proof.

$$\begin{aligned} M^\alpha \circ D^\beta \circ \mathcal{F} &= M^\alpha((-1)^{|\beta|} \mathcal{F} \circ M^\beta) \quad \text{by (ii)} \\ &= (-1)^{|\beta|} (M^\alpha \circ \mathcal{F}) \circ M^\beta \\ &= (-1)^{|\beta|} (\mathcal{F} \circ D^\alpha) \circ M^\beta \quad \text{by (i)}. \end{aligned}$$

i.e.,

$$\xi^\alpha D_\xi^\beta \mathcal{F}(f)(\xi) = (-1)^{|\beta|} \mathcal{F}(D^\alpha x^\beta f)(\xi).$$

Since $f \in \mathcal{S}$, by Proposition 2.2(ii) and Proposition 2.3, $D^\alpha x^\beta f \in \mathcal{S} \subset L^1$. By Riemann-Lebesgue lemma, $\mathcal{F}(D^\alpha x^\beta f) \in C_0$, showing the left hand side of the above equation is bounded. f is in C^∞ for the same reason using Theorem 2.6(ii). Hence $\mathcal{F}f \in \mathcal{S}$. $\square$

3 The main theorem

Theorem 3.1. *The Fourier transform $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$ is a bijective map, where the inverse $\mathcal{F}^{-1}$ (Fourier inversion) is given by:*

$$\mathcal{F}^{-1}(f)(x) := \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} f(\xi) e^{ix \cdot \xi} d\xi.$$

Proof. Proofs of the two Claims 4.1, 4.2 will be given in the next section.

$$\begin{aligned} \mathcal{F}^{-1} \circ \mathcal{F}(f)(x) &= \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} \mathcal{F}(f)(\xi) e^{ix \cdot \xi} d\xi \\ &= \lim_{\epsilon \rightarrow 0^+} \int_{\mathbb{R}^n} \mathcal{F}(f)(\xi) \frac{1}{(2\pi)^{n/2}} e^{-\epsilon|\xi|^2} \cdot e^{ix \cdot \xi} d\xi \quad \text{by DCT} \\ &= \lim_{\epsilon \rightarrow 0^+} \int_{\mathbb{R}^n} \hat{f}(\xi) g(\xi) d\xi, \quad \text{where } g(\xi) := \frac{1}{(2\pi)^{n/2}} e^{-\epsilon|\xi|^2 + ix \cdot \xi} \\ &= \lim_{\epsilon \rightarrow 0^+} \int_{\mathbb{R}^n} f(y) \hat{g}(y) dy \quad \text{by Lemma 1.4} \\ &= \lim_{\epsilon \rightarrow 0^+} \int_{\mathbb{R}^n} f(y) \frac{1}{(4\pi\epsilon)^{n/2}} e^{-|x-y|^2/4\epsilon} dy \quad \text{by my Claim 4.1} \\ &= \lim_{\epsilon \rightarrow 0^+} \int_{\mathbb{R}^n} f(y) K_\epsilon(x-y) dy, \quad \text{where } K_\epsilon(x) := \frac{1}{(4\pi\epsilon)^{n/2}} e^{-|x|^2/4\epsilon} \\ &= f(x) \quad \text{by my Claim 4.2} \end{aligned}$$

Checking $\mathcal{F} \circ \mathcal{F}^{-1} = Id$ is similar. □

Corollary 3.2. *If $f, g \in \mathcal{S}$, then $\langle \mathcal{F}f, \mathcal{F}g \rangle_{L^2} = \langle f, g \rangle_{L^2} := \int_{\mathbb{R}^n} f \bar{g}$.*

Proof. If $f, g \in \mathcal{S}$, then

$$\begin{aligned} \langle \mathcal{F}(f), \mathcal{F}(g) \rangle_{L^2} &= \int \mathcal{F}(f) \cdot \overline{\mathcal{F}(g)} = \int \mathcal{F}(f) \cdot \mathcal{F}^{-1}(\bar{g}) \\ &= \int f \cdot \mathcal{F}\mathcal{F}^{-1}(\bar{g}) \quad \text{by Lemma 1.4} \\ &= \int f \bar{g} = \langle f, g \rangle_{L^2} \end{aligned}$$

□

Corollary 3.3 (Plancherel). *$\mathcal{F}$ extends (by density of $\mathcal{S}$ in L^2) to a unitary operator $\mathcal{F} : L^2 \rightarrow L^2$, i.e.,*

$$\langle \mathcal{F}f, \mathcal{F}g \rangle_{L^2} = \langle f, g \rangle_{L^2} \quad \forall f, g \in L^2.$$

Proof. We have for all $f \in \mathcal{S}$, $\|\mathcal{F}f\|_{L^2} = \|f\|_{L^2}$. By Corollary 2.4, $\mathcal{S}$ is dense in L^2 . □

4 The two claims

Claim 4.1. *Given $x \in \mathbb{R}^n$ and $\epsilon > 0$, if*

$$g(\xi) := \frac{1}{(2\pi)^{n/2}} e^{-\epsilon|\xi|^2 + ix \cdot \xi},$$

then

$$\hat{g}(y) = \frac{1}{(4\pi\epsilon)^{n/2}} e^{-|x-y|^2/4\epsilon}.$$

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Proof. We simply compute $\hat{g}(y)$. It's equivalent to showing

$$\frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{-\epsilon|\xi|^2 + i(x-y)\cdot\xi} d\xi = \frac{1}{(4\pi\epsilon)^{n/2}} e^{-|x-y|^2/4\epsilon},$$

which can be reduced to (by Fubini) showing that for each $j = 1, 2, \dots, n$.

$$\frac{1}{2\pi} \int_{\mathbb{R}} e^{-\epsilon\xi_j^2 + i(x_j - y_j)\xi_j} d\xi_j = \frac{1}{(4\pi\epsilon)^{1/2}} e^{-(x_j - y_j)^2/4\epsilon},$$

which can be reduced to (by change of variables $\xi := x_j - y_j$ and $\xi_j := x$) showing that

$$\frac{1}{2\pi} \int_{\mathbb{R}} e^{-\epsilon x^2 - i\epsilon x} dx = \frac{1}{(4\pi\epsilon)^{1/2}} e^{-\xi^2/4\epsilon},$$

which can be reduced to showing that given an $\epsilon > 0$, then $\forall \xi \in \mathbb{R}$,

$$e^{\xi^2/4\epsilon} \cdot \mathcal{F}(f)(\xi) = \frac{1}{(2\epsilon)^{1/2}}, \text{ where } f(x) := e^{-\epsilon x^2} \in \mathcal{S} \quad (4.1)$$

To show (4.1) holds, we will first show $D_\xi(e^{\xi^2/4\epsilon} \cdot \mathcal{F}(f)(\xi)) = 0$, i.e. LHS of (4.1) is a constant in ξ .

We compute

$$D_x f = \frac{1}{i} \frac{d}{dx} (e^{-\epsilon x^2}) = \frac{1}{i} \cdot (-2\epsilon x) e^{-\epsilon x^2} = \frac{-2\epsilon}{i} M(f) \quad (4.2)$$

Thus

$$\begin{aligned} D_\xi \circ \mathcal{F}(f) &= (-1) \mathcal{F} \circ M(f) \quad \text{by Theorem 2.6} \\ &= (-1) \mathcal{F} \left(\frac{i}{-2\epsilon} D_x(f) \right) = \frac{i}{2\epsilon} \mathcal{F} \circ D_x(f) \text{ by (4.2)} \\ &= \frac{i}{2\epsilon} M \circ \mathcal{F}(f) = \frac{i}{2\epsilon} \xi \cdot \mathcal{F}(f) \quad \text{by Theorem 2.6} \end{aligned} \quad (4.3)$$

Now we can compute, by product rule, that,

$$\begin{aligned} D_\xi(e^{\xi^2/4\epsilon} \cdot \mathcal{F}(f)(\xi)) &= D_\xi(e^{\xi^2/4\epsilon}) \cdot \mathcal{F}(f) + e^{\xi^2/4\epsilon} \cdot D_\xi \circ \mathcal{F}(f) \\ &= \frac{1}{i} \cdot \frac{2\xi}{4\epsilon} e^{\xi^2/4\epsilon} \cdot \mathcal{F}(f) + e^{\xi^2/4\epsilon} \cdot \frac{i}{2\epsilon} \xi \cdot \mathcal{F}(f) = 0 \text{ by (4.3)} \end{aligned}$$

Thus $e^{\xi^2/4\epsilon} \cdot \mathcal{F}(f)(\xi)$ is a constant in ξ . To know the value of this constant, we evaluate at $\xi = 0$, then

$$e^{\xi^2/4\epsilon} \cdot \mathcal{F}(f)(\xi) = e^{0^2/4\epsilon} \cdot \mathcal{F}(f)(0) = \frac{1}{(2\pi)^{1/2}} \int_{\mathbb{R}} e^{-\epsilon x^2} dx = \frac{1}{(2\epsilon)^{1/2}}.$$

□

Claim 4.2. If $f \in \mathcal{S}$, then for each $x \in \mathbb{R}^n$,

$$\lim_{\epsilon \rightarrow 0^+} \int_{\mathbb{R}^n} f(y) K_\epsilon(x - y) dy = \lim_{\epsilon \rightarrow 0^+} \int_{\mathbb{R}^n} f(x - y) K_\epsilon(y) dy = f(x), \quad (4.4)$$

where

$$K_\epsilon(x) := \frac{1}{(4\pi\epsilon)^{n/2}} e^{-|x|^2/4\epsilon}. \quad (4.5)$$

Proof. We follow Chapter 2.4 of Real Analysis by Stein and Shakarchi and show K_ϵ is a good kernel. Namely,

- (i) $\int_{\mathbb{R}^n} K_\epsilon(y) dy = 1$, for all $\epsilon > 0$.
- (ii) For each $\eta > 0$, $\int_{|y| > \eta} K_\epsilon(y) dy \rightarrow 0$ as $\epsilon \rightarrow 0^+$.

(i) and (ii) hold by change of variables $y = x/\sqrt{\epsilon}$.

By continuity of f at x , for each $\epsilon' > 0$, there is an $\eta = \eta(\epsilon') > 0$ such that

$$|y| < \eta \quad \text{implies} \quad |f(x - y) - f(x)| < \epsilon'. \quad (4.6)$$

We compute, with this choice of $\eta > 0$,

$$\begin{aligned} \left| \int_{\mathbb{R}^n} f(x - y) K_\epsilon(y) dy - f(x) \right| &\leq \int_{\mathbb{R}^n} |f(x - y) - f(x)| \cdot K_\epsilon(y) dy \quad \text{Since } K_\epsilon > 0 \\ &= \int_{|y| > \eta} |f(x - y) - f(x)| \cdot K_\epsilon(y) dy + \int_{|y| \leq \eta} |f(x - y) - f(x)| \cdot K_\epsilon(y) dy \\ &\leq \int_{|y| > \eta} 2 \sup_{\mathbb{R}^n} |f| \cdot K_\epsilon(y) dy + \int_{|y| < \eta} \epsilon' \cdot K_\epsilon(y) dy \quad \text{by (4.6)} \\ &\leq 2 \sup_{\mathbb{R}^n} |f| \cdot \int_{|y| > \eta} K_\epsilon(y) dy + \epsilon' \cdot 1 \end{aligned}$$

Letting $\epsilon' \rightarrow 0$ and letting $\epsilon \rightarrow 0$, using (ii), we've proved (4.4).

□